Theoretical Physics 2: Quantum Mechanics

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Summer Semester 2026

Inhaltsverzeichnis

1 Preface

1.1 Purpose of This Script

This script is designed as a learning-oriented companion to the course *Theoretical Physics 2: Quantum Mechanics*. It is not meant to replace the lecture notes, exercise sheets, or official course material. Its purpose is to organize the material into a coherent sequence of concepts, derivations, examples, and exam-oriented problem patterns.

The guiding principle is simple: every formula should be connected to a physical interpretation, every formal statement should be connected to a calculational method, and every standard calculation should be connected to typical exam questions.

How to use this script

Read each chapter in three passes.

1. First, read the definitions and conceptual explanations to understand what the objects mean.
2. Second, work through the derivations and examples with pen and paper.
3. Third, use the exam checklist at the end of the chapter to test whether you can reproduce the central arguments without looking them up.

1.2 Main Sources

The main source for this script is the official lecture material for *Theoretical Physics 2: Quantum Theory*. The handwritten lecture notes are used as a supplementary source whenever they clarify the order of presentation or contain useful intermediate calculations. The problem sheets and their solutions are used primarily to identify exam-relevant techniques and recurring calculation patterns.

The source priority used in this script is:

1. official exam information, exam sheets, topic lists, and learning goals;
2. official lecture slides or lecture script;
3. exercise sheets and official sample solutions;
4. personal submitted solutions, only when they are readable and physically consistent;
5. supplementary textbooks.

Source-critical use of solutions

Personal solutions and sample solutions are not copied blindly. They are used as orientation for notation, level of detail, and typical problem-solving strategies. If a derivation appears incomplete, ambiguous, or inconsistent, the script gives a clean derivation instead.

1.3 Recommended Literature

The following books are listed in or are naturally aligned with the course material. They are useful in different ways:

- F. Schwabl, *Quantenmechanik (QM I): Eine Einfuehrung*. Useful as a German-language reference with a structure close to many theoretical physics lectures.
- C. Cohen-Tannoudji, B. Diu, and F. Laloe, *Quantum Mechanics*, Volumes 1 and 2. Very detailed and especially useful for conceptual explanations, examples, and complementary material.
- G. Grawert, *Quantenmechanik*. A compact German-language study text.
- J. J. Sakurai, *Modern Quantum Mechanics*. Particularly useful for the abstract Hilbert-space formalism, Dirac notation, spin, and angular momentum.
- A. Galindo and P. Pascual, *Quantum Mechanics I and II*. A more advanced and mathematically careful reference.

1.4 Structure of the Script

The script is organized so that the formalism is built gradually. The early chapters introduce wave functions, Hilbert spaces, probability interpretation, and the Schrödinger equation. The middle chapters develop operators, commutators, measurements, uncertainty relations, and standard one-dimensional systems. Later chapters cover angular momentum, spin, perturbation theory, identical particles, and elementary quantum information concepts such as qubits, Bloch vectors, and entanglement.

No.	File	Topic
00	`00_preface.tex`	Purpose, sources, learning str
01	`01_wave_functions_and_statistical_interpretation.tex`	Wave functions, Hilbert space tation
02	`02_schroedinger_equation_and_time_evolution.tex`	Schrödinger equation, time e current
03	`03_free_particles_wave_packets_and_fourier_representation.tex`	Free particles, plane waves, F spreading
04	`04_one_dimensional_potentials_and_tunneling.tex`	One-dimensional potentials, scattering
05	`05_delta_potential_bound_states_and_discontinuities.tex`	Delta potential, bound states
06	`06_operators_commutators_and_ehrenfest_theorem.tex`	Operators, commutators, Ehr
07	`07_measurements_observables_and_uncertainty.tex`	Measurements, observables, ments, uncertainty
08	`08_dirac_notation_and_abstract_hilbert_space.tex`	Dirac notation, projectors, m sentations
09	`09_harmonic_oscillator_and_ladder_operators.tex`	Harmonic oscillator and ladd
10	`10_coherent_states_and_quasiclassical_motion.tex`	Coherent states and quasicals
11	`11_angular_momentum_and_spherical_harmonics.tex`	Orbital angular momentum nics
12	`12_central_potentials_and_hydrogen_atom.tex`	Central potentials and the hy
13	`13_scattering_theory_partial_waves_and_resonances.tex`	Scattering theory, partial wav
14	`14_spin_half_systems_and_pauli_matrices.tex`	Spin-1/2 systems and Pauli m
15	`15_time_dependent_spin_dynamics_and_rabi_formula.tex`	Rotating frame, magnetic res
16	`16_addition_of_angular_momenta_and_spin_coupling.tex`	Addition of angular momenta
17	`17_time_independent_perturbation_theory.tex`	Non-degenerate and degenera ry
18	`18_identical_particles_bosons_fermions_and_symmetrization.tex`	Identical particles, bosons, fer on
19	`19_qubits_bloch_sphere_and_entanglement.tex`	Qubits, Bloch sphere, Bell sta
20	`20_exam_strategy_formula_sheet_and_problem_patterns.tex`	Exam strategy, formula sheet

1.5 Notation and Conventions

Unless stated otherwise, states are denoted either as wave functions $\psi(x)$, $\psi(\vec{x})$, or as abstract vectors $|\psi\rangle$. Operators are denoted by hats when helpful, for example $\hat{H}$, $\hat{X}$, and $\hat{P}$. The inner product may be written either as an integral in position space or in Dirac notation:

$$\langle\phi|\psi\rangle = \int \mathrm{d}x\, \phi^*(x)\psi(x). \tag{1.1}$$

For a single particle in one spatial dimension, the canonical position and momentum operators are usually represented by

$$(\hat{X}\psi)(x) = x\psi(x), \qquad (\hat{P}\psi)(x) = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}\psi(x), \tag{1.2}$$

and they satisfy the canonical commutation relation

$$\left[\hat{X}, \hat{P}\right] = \mathrm{i}\hbar\mathbb{1}. \tag{1.3}$$

Notation matters

Many mistakes in quantum mechanics are notation mistakes. Always distinguish between a state, a wave function, an operator, an eigenvalue, and a measurement result. For example, $\psi(x)$ is a representation of a state, while $|\psi\rangle$ is the abstract state itself.

1.6 How to Study Problem Sheets

The problem sheets are not merely exercises after the theory. They define the calculational core of the course. For each problem, the following questions are useful:

1. Which postulate or principle is being tested?
2. Which representation is most convenient: position space, momentum space, matrix notation, or Dirac notation?
3. Which boundary conditions or normalization conditions are required?
4. Which quantities are physically observable?
5. Which limiting cases should be checked?

Typical quantum-mechanical workflow

A large fraction of problems follows the same pattern:

1. Identify the Hilbert space and the relevant operator.
2. Solve the eigenvalue problem or expand the state in a useful basis.
3. Normalize states and impose boundary conditions.
4. Compute probabilities, expectation values, or transition amplitudes.
5. Interpret the result physically and check limiting cases.

1.7 Common Sources of Errors

- Confusing ψ with $|\psi|^2$. The wave function is a complex amplitude; only the modulus squared is a probability density.
- Forgetting normalization factors in continuous bases, especially in Fourier transforms.

- Treating non-normalizable plane waves as physical states without using wave packets or generalized normalization.
- Applying boundary conditions only to ψ, but forgetting the derivative condition when the potential is finite.
- Applying derivative continuity at a delta potential, where the derivative has a jump instead.
- Ignoring degeneracy when using perturbation theory.
- Mixing up active time evolution of states with passive changes of basis.

1.8 Summary

This chapter sets the purpose and structure of the script. The script is based primarily on the official lecture material and the exercise sheets, with handwritten notes and personal solutions used only as supplementary material. The goal is to produce a coherent learning document that connects concepts, derivations, examples, and exam-style problem solving.

1.9 Exam Checklist

Before moving on, make sure that you can answer the following questions:

- What is the purpose of this script, and how should it be used?
- Which sources have priority when lecture notes, exercises, and personal solutions differ?
- What is the difference between a state, a wave function, an operator, and an observable?
- Why are normalization, boundary conditions, and basis choice central in quantum-mechanical calculations?
- Which chapters are likely to be especially relevant for exam-style calculations?